

The Dyadic Shadow Equals a Local Oscillation Obligation

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Abstract

Every prefix inside a dyadic block is the left block-boundary prefix plus one local consecutive-block increment. Consequently TPC-159 boundary control closes the full endpoint target exactly when a named-atom local oscillation bound is added. No averaging or exceptional-endpoint deletion is permitted. The classification is REDUCTION_L1; no L2 arithmetic progress or endpoint credit is claimed.

1 Frozen target

The selected route is `0161.bad_endpoint_pointwise_fixed_atom` [1]. The required signature is actual fixed- h_0 packet, named fixed atom, deterministic every prefix, deterministic every scale, fixed- X power at that atom, and actual active support. Throughout $h_0 = 2$ is only a source-backed data fact.

2 Exact result

Theorem 1. *For a block boundary b and $b < k \leq b'$, $|S_k| \leq |S_b| + \max_{b < r \leq b'} |S_r - S_b|$. Therefore the missing dyadic-shadow estimate is precisely a local maximal increment theorem at the same fixed atom.*

Proof. For $b < k \leq b'$, $S_k = S_b + (S_k - S_b)$, giving full-prefix control from boundary and local-increment control. Conversely each increment is $S_k - S_b$, so it is at most twice a full-prefix maximum. Thus the two maxima are equivalent up to factor two on the identical endpoint family; constant factors do not alter a fixed power. $\square$

The smallest missing literal theorem is `FIXED_ATOM_LOCAL_MAXIMAL_INCREMENT_POWER_SAVING`.

3 Scoped stop and claim firewall

No new scoped stop is declared in this paper.

The result is L0/L1 only. Phase averages and Lebesgue-almost-every phase are not named atoms. Fixed $h_0 = 2$ is not decay. Archive addressing is not canonical physical representation, and empty-domain truth is not actual active support. The named-atom endpoint charge is zero, so the strict $1/400$ budget is unpaid. We claim no program-positive L2, prime-pair lower bound, or twin-prime theorem.

4 Reproducibility

The adjacent JSON payload records the full six-axis signature, source lock, typed result, exact scoped-stop cell, endpoint ledger, and claim firewall. The audit artifact contains eight true mutation-rejection assertions.

References

- [1] Liang Wang. Mvp8 source-phase route decision, 2026.